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Mounting device for a fall protection device and methods of using the same

Abstract

Various embodiments are directed to mounting devices for fall protection devices and methods of using the same. In various embodiments, a mounting device may comprise a mounting bracket configured for attachment to a materials handling vehicle; a rotary arm rotatably attached to the mounting bracket at an arm base, wherein the rotary arm is configured to rotate relative to the mounting bracket about an axis of rotation defined at the arm base, and wherein the rotary arm defines a webbing retention feature configured for receiving a webbing of a fall protection device to define a dynamic engagement of the rotary arm with an intermediate webbing portion defined along the webbing; a sensing device configured to capture sensor data associated with the intermediate webbing portion; and a controller configured to detect a connection condition associated with the fall protection device based on the sensor data captured by the sensing device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4942943	12/1989	Flaherty	N/A	N/A
11459780	12/2021	Martin Iglesias	N/A	E04G 21/3261
12102851	12/2023	Strauss	N/A	A62B 35/0068
2012/0325586	12/2011	Meggs et al.	N/A	N/A
2013/0146699	12/2012	O'Brien et al.	N/A	N/A
2015/0375020	12/2014	Palet	N/A	N/A
2017/0368387	12/2016	Fife et al.	N/A	N/A
2018/0126198	12/2017	Troy et al.	N/A	N/A
2020/0096952	12/2019	Hu et al.	N/A	N/A
2022/0134149	12/2021	Carlson et al.	N/A	N/A
2022/0195742	12/2021	Fife	N/A	N/A
2023/0127292	12/2022	Palet	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202822515	12/2012	CN	N/A
113508426	12/2020	CN	N/A
2002-104776	12/2001	JP	N/A
2005-061106	12/2004	JP	N/A
2019-188009	12/2018	JP	N/A
2020/194121	12/2019	WO	N/A
2021/099874	12/2020	WO	N/A

OTHER PUBLICATIONS

Capital Safety, “Standing Seam Swiveling Roof Anchor”, Protecta Product Sheet, 2 pages, 2011. cited by applicant
English Translation of JP Notice of Allowance, including Search Report dated Oct. 16, 2024 for JP Application No. 2023117538, 3 page(s). cited by applicant
Extended European Search Report Mailed on Nov. 28, 2023 for EP Application No. 23185247, 9 page(s). cited by applicant
Harness Land, “Standing Seam Roof Anchor Clamp”, 3 pages, retrieved from the Internet at <URL: https://www.harnessland.com/Standing-Seam-Roof-Anchor-p/00250.htm> on Jan. 11, 2023. cited by applicant

Honeywell, “Miller DuraHoistTM Safety Systems”, 8 pages, 2017. cited by applicant
JP Notice of Allowance, including Search Report Mailed on Oct. 16, 2024 for JP Application No. 2023117538, 3 page(s). cited by applicant
JP Office Action Mailed on May 2, 2024 for JP Application No. 2023117538, 7 page(s). cited by applicant
Non-Final Rejection Mailed on Oct. 17, 2023 for U.S. Appl. No. 17/813,869, 9 page(s). cited by applicant
Notice of Allowance and Fees Due (PTOL-85) Mailed on Mar. 13, 2024 for U.S. Appl. No. 17/813,869, 7 page(s). cited by applicant
U.S. Appl. No. 17/813,869, filed Jul. 20, 2022, U.S. Pat. No. 12,070,635, Patented. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/813,869, entitled “MOUNTING DEVICE FOR A FALL PROTECTION DEVICE AND METHODS OF USING THE SAME” and filed on Jul. 20, 2022, which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION
(1) Various embodiments described herein relate generally to safety equipment or personal protective equipment (PPE), including mounting devices and fall protection devices used by operators within a materials handling environment during operation of various materials handling vehicles capable of operating at an elevated height.
BACKGROUND
(2) Generally, in material handling environments like, but not limited to, high volume distribution and fulfillment operations, materials handling vehicles can be used to retrieve objects from various storage locations defined at elevated heights. Fall protection equipment may be utilized by an operator of a materials handling vehicle to minimize risk of falling from the elevated height during operation of a materials handling vehicles. Applicant has identified several technical challenges associated with utilizing fall protection devices during operation of materials handling vehicles and other associated systems and methods. Through applied effort, ingenuity, and innovation, many of these identified challenges have been overcome by developing solutions that are included in embodiments of the present invention, many examples of which are described in detail herein.
BRIEF SUMMARY
(3) Various embodiments are directed to a mounting device for a fall protection device and methods of using the same. In various embodiments, a mounting device may comprise a mounting bracket configured for attachment to a materials handling vehicle; a rotary arm rotatably attached to the mounting bracket at an arm base, wherein the rotary arm is configured to rotate relative to the mounting bracket about an axis of rotation defined at the arm base, and wherein the rotary arm defines a webbing retention feature configured for receiving a webbing of a fall protection device to define a dynamic engagement of the rotary arm with an intermediate webbing portion defined along a length of the webbing; a first sensing device configured to capture first sensor data associated with the intermediate webbing portion; and a controller configured to detect a connection condition associated with the fall protection device based at least in part on the first

sensor data captured by the first sensing device.

(4) In various embodiments, the webbing retention feature may define an extended webbing engagement point at which the rotary arm is configured to contact the intermediate webbing portion to at least partially define an arrangement of the webbing relative to the mounting device. In certain embodiments, the rotary arm may comprise an extension arm that is rigidly secured to the arm base at a proximal arm end, the extension arm defining an arm length that extends from the proximal arm end in an outward radial direction to a distal arm end, wherein the outward radial direction being defined relative to the axis of rotation. In certain embodiments, the webbing retention feature may define at a distal position along the extension arm, and wherein a rotation of the rotary arm about the axis of rotation causes a corresponding rotation of the extended webbing engagement point about the axis of rotation such that rotary arm is configured to facilitate an extended range of motion for an operator relative to the fall protection device. In various embodiments, the rotary arm may be configured for coupling with the fall protection device such that a body of the fall protection is secured relative to the rotary arm, the rotary arm being configured for engagement with the fall protection device such that a rotation of the rotary arm about the axis of rotation causes a corresponding rotation of the fall protection device through a corresponding range of rotational motion. In certain embodiments, the axis of rotation may be defined by a central axis of the arm base, and wherein the rotary arm is configured for securing the body of the fall protection device relative to the arm base of the rotary arm such that the axis of rotation is defined at both the arm base and the fall protection device secured thereto.

(5) In various embodiments, the webbing retention feature may define an opening configured for the webbing of the fall protection device to be threaded therethrough such that at least a portion of the intermediate webbing portion is disposed within the opening. In various embodiments, the rotary arm may be rotatably attached to an arm interface portion of the mounting bracket defining a downward-facing bottom surface such that the rotary arm is configured to define a position vertically below the mounting bracket upon the mounting bracket being secured relative to the materials handling vehicle, and wherein the axis of rotation is defined in a direction at least substantially perpendicular to the downward-facing bottom surface. In various embodiments, the rotary arm may be rotatably attached to the mounting bracket via one or more fastening elements comprising a slip ring. In various embodiments, the first sensor data captured by the first sensing device may be configured to facilitate a detection of a movement condition defined by one or more movements of the intermediate webbing portion relative to the webbing retention feature. In certain embodiments, the first sensing device may comprise an imaging device and the first sensor data is defined by imaging data comprising at least one image of the intermediate webbing portion. In certain embodiments, the first sensing device may be positioned at the webbing retention feature.

(6) In various embodiments, the mounting device may further comprise a second sensing device configured to capture second sensor data to facilitate a detection of a movement condition of the mounting device. In various embodiments, the second sensor data captured by the second sensing device may correspond at least in part to a movement of one or more of the mounting bracket and the rotary arm, and wherein the controller is configured to detect the movement condition of the mounting device based at least in part on the second sensor data captured by the second sensing device. In certain embodiments, the second sensor data may be positioned relative to the mounting bracket such that the second sensor data captured by the second sensing device corresponds to one or more of a linear movement of the mounting bracket and a rotational movement of the rotary arm. In certain embodiments, the second sensing device may comprise a motion-sensing device comprising a gyroscope and an accelerometer, the motion-sensing device being configured to execute a six-degrees-of-freedom motion-sensing operation.

(7) In various embodiments, the connection condition associated with the fall protection device may be defined by a detection of a connected configuration of the fall protection device relative to an operator attachment operatively secured to an operator. In various embodiments, the controller

may be configured to detect the connected configuration associated with fall protection device based on a detected first movement condition defined by the intermediate webbing portion disposed at the webbing retention feature. In certain embodiments, the controller may be configured to detect the connected configuration associated with fall protection device further based on a detected second movement condition defined by a detected movement of the mounting device. Further, in certain embodiments, the controller may be configured to detect the connected configuration associated with fall protection device further based on a drive signal received by the controller, the drive signal corresponding to a user-initiated operation of the materials handling vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. **1** illustrates a perspective view of a materials handling vehicle and an exemplary mounting device associated with fall protection equipment according to an example embodiment described herein;

(3) FIGS. **2**A and **2**B illustrate various views of an exemplary mounting device associated with fall protection equipment according to an example embodiment described herein;

(4) FIG. **3** illustrates a rear view of a materials handling vehicle and an exemplary mounting device associated with fall protection equipment according to an example embodiment described herein; and

(5) FIG. **4** illustrates a schematic view of an exemplary apparatus in accordance with various embodiments.

DETAILED DESCRIPTION

(6) The present disclosure more fully describes various embodiments with reference to the accompanying drawings. It should be understood that some, but not all embodiments are shown and described herein. Indeed, the embodiments may take many different forms, and accordingly this disclosure should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.

(7) It should be understood at the outset that although illustrative implementations of one or more aspects are illustrated below, the disclosed assemblies, systems, and methods may be implemented using any number of techniques, whether currently known or not yet in existence. The disclosure should in no way be limited to the illustrative implementations, drawings, and techniques illustrated below, but may be modified within the scope of the appended claims along with their full scope of equivalents. While values for dimensions of various elements are disclosed, the drawings may not be to scale.

(8) The words “example,” or “exemplary,” when used herein, are intended to mean “serving as an example, instance, or illustration.” Any implementation described herein as an “example” or “exemplary embodiment” is not necessarily preferred or advantageous over other implementations.

(9) Picking or workstations are essential components of high volume distribution and fulfillment operations. Conventionally, order picking requires an order picker to take an order list, walk through racks of products filled with containers of products to pick from, picking the listed products from product containers, and placing the picked products into an order container for delivery to packaging. However, this solution is slow and requires intensive manpower. Thus, automated picking or workstations are used in more recent systems. For example, various materials handling vehicles comprising order pickers may be configured to move throughout a materials

handling environment, a storage environment, and/or the like to retrieve objects from storage locations defined at various heights. Where such materials handling vehicles can be configured to raise a platform upon which an operator stands in an operating position during operation of the vehicle to an elevated height corresponding to the storage location, various industry safety standards have been developed that require the use of fall protection equipment configured to facilitate a secured connection of the operator (e.g., a harness being worn by the operator) to an anchored fastener on the materials handling vehicle. Often, however, installing the fall protection equipment in a fully installed configuration relative to both the operator and the picker may be a cumbersome process for an operator that requires the operator to spend time executing a number of installation steps prior to operating the picker, which may represent a costly loss of operation time in high volume distribution and fulfillment operations. As such, picker operators frequently forego using the required fall protection equipment, which can create an extremely dangerous operating condition with a high risk of serious operator injury, and may result in industry sanctions, regulatory punishments, and/or monetary damages based on a lack of compliance with established safety standards.

(10) Various embodiments described herein are directed to mounting devices for securing a fall protection device relative to a materials handling vehicle that utilizes captured sensor data to determine whether the fall protection device is properly installed in a connected configuration relative to an operator during operation of the materials handling vehicle. As described herein, the present invention includes a mounting bracket configured for attachment to a materials handling vehicle and a rotary arm rotatably attached to the mounting bracket that is configured for coupling with the fall protection device in order to secure the fall protection device relative to the mounting device. The rotary arm of the present invention defines a webbing retention element at a distal end thereof that at which the rotary arm is configured to receive a portion of the webbing connecting the body of the fall protection device to an operator standing therebelow. Accordingly, the mounting device is configured such that a rotation of the rotary arm enables an extended range of motion for the operator as the operator moves relative to the mounted body of the fall protection device throughout the workspace. Further, the engagement of the webbing of the fall protection device with the rotary arm, as described herein, causes the drag forces present within the webbing during operator movement to be at least substantially minimized such that the operator experiences a minimal amount of resistance from the fall protection device that is anchoring the operator to the mounting device.

(11) Further, the present invention includes a sensing devices configured to capture sensor data associated with the movement the mounting device within an environment and the movement (e.g., micro-movement) of the webbing portion engaged with the rotary arm relative to the webbing retention feature. The mounting device incudes a controller configured to detect a connection condition associated with the fall protection device based at least in part on the sensor data captured by the sensing device(s).

(12) Accordingly, as described herein, the present invention promotes user comfort and ease-of accessibility throughout a workspace defined relative to a materials handling vehicle, while drastically reducing the risk of a materials handling vehicle being deliberately and/or unintentionally operated in a dangerous and/or non-compliant manner.

(13) FIG. **1** illustrates a perspective view of a materials handling vehicle having an exemplary mounting device for detecting a connection condition associated with fall protection device. In particular, FIG. **1** illustrates a perspective view of an exemplary mounting device for a fall protection device configured for operatively connecting a fall protection device to a materials handling vehicle and detecting a connection condition associated with the fall protection device relative to an operator coupling element (e.g., at an operator-worn harness). For example, an exemplary mounting device may be configured to detect whether a fall protection device connected thereto is installed in a fully installed configuration relative to an operator coupling element to

determine whether an operator present within a workspace defined by the materials handling vehicle is sufficiently secured to the fall protection device during operation of the materials handling vehicle.

(14) In various embodiments, for example, a mounting device **100** may be configured to detect a connection condition associated with a fall protection device **20** configured for use within a workspace defined by a materials handling vehicle **10**. For example, a materials handling vehicle **10**, such as, for example, an order picker, may be used by a human operator **1** within a materials handling environment (e.g., a storage warehouse) to facilitate the retrieval and/or storage of an object to and/or from a storage location located at an elevated height above the ground floor. An exemplary materials handling vehicle **10** may comprise a lift configured to be selectively moveable in a vertical direction to a plurality of vertical heights in order to facilitate the storage and/or retrieval of an object from an elevated storage location. The lift of the materials handling vehicle **10** may comprise a lift platform **11***a* upon which an operator **1** may stand as the lift is moved between various vertical heights in order to enable the operator to assist in the storage and/or retrieval operation from the elevated storage location. In various embodiments, the materials handling vehicle **10** comprises a workstation **11** at which the operator **1** may stand in an operating position in order to operate (e.g., drive, lift, and/or otherwise control) the materials handling vehicle **10**. In various embodiments, as illustrated, the workstation **11** of the materials handling vehicle **10** may be defined on the lift platform **11***a*, such that the operating position of the materials handling vehicle **10** is defined by the operator **1** standing on the lift platform **11***a* in a front-facing position towards the vehicle operation controls (e.g., user controls accessible to an operator **1** standing in the operating position that are configured to facilitate operation of the drive assembly and the lift assembly of the materials handling vehicle **10**).

(15) To maintain compliance with various safety standards, protocols, and/or the like designed mitigate the risk of personal injury associated with an operator working at an elevated height, an operator **1** may use one or more items of personal protective equipment (PPE) defining fall protection equipment (FPE), such as, for example, a fall protection device **20** and a wearable harness **30**. In various embodiments, a fall protection device **20** (e.g., a personal fall limiter) may be used to operably secure an operator **1** wearing a harness **30** relative to one or more anchors points. For example, as illustrated, a fall protection device **20** may be configured to operably attach the operator **1** to an exemplary mounting device **100** that is secured to an upper frame element **12** (e.g., a rail, a beam, and/or the like) of the materials handling vehicle **10** that positioned above the workspace **11** defined by the materials handling vehicle **10**, such as, for example, on a top portion of the lift that is configured to move vertically with the platform **11***a* such that both the upper frame element **12** and the mounting device **100** secured thereto maintain a position above the operator **1** throughout operation of the materials handling vehicle **10**.

(16) An exemplary mounting device **100** may be configured to receive a fall protection device **20** such that the fall protection device **20** may be coupled to the mounting device **100** to facilitate the mounting of the fall protection device **20** relative to the upper frame element **12** of the materials handling vehicle **10**. For example, the mounting device **100** may be configured to facilitate the anchoring of an operator **1** relative to the upper frame element **12** of the materials handling vehicle **10** via the fall protection device **20** that is mounted to the mounting device **100**. As an illustrative example, FIG. **1** shows an exemplary fall protection device **20** mounted to an exemplary mounting device **100** and in an installed configuration relative to an operator harness **30** (e.g., an operator attachment **31**) being worn by an operator **1** standing in an operating position. In various embodiments, the installed configuration of a fall protection device **20** that is mounted to an exemplary mounting device **100** may be defined by a connection of the attachment element **22** of the fall protection device **20** to the operator attachment **31** of the harness **30**.

(17) It should be understood that many types and configurations of safety/fall harnesses are known in the PPE and FPE industry, including full body harnesses and partial or hip/waist fall harnesses,

all, or most, of which are suitable for use with the concepts disclosed herein. Accordingly, the wearable safety harness **30** depicted in FIG. **1** is provided for purposes of illustration and further specific details of the harness **30** will not be discussed herein except for those required for an understanding of the disclosed concepts, and that the appended claims are not limited to any specific details of a harness unless expressly recited in the claims.

(18) In various embodiments, a mounting device may be installed at materials handling vehicle relative to a workspace and configured to detect a connection condition defined at least in part by a connection of a fall protection device relative to an operator (e.g., an operator attachment of a wearable harness). For example, as illustrated in FIG. **1**, an exemplary mounting device **100** may be installed at an upper frame element **12** of an exemplary materials handling vehicle **10** and configured to detect a connection condition of a fall protection device **20** relative to an operator **1** present within the workspace **11**, wherein the connection condition is defined at least in part by the connected configuration of the attachment element **22** of the fall protection device **20** mounted to the mounting device **100** relative to an operator attachment **31** defined by a wearable harness **30** being worn by the operator **1**. In various embodiments, the operator attachment **31** may comprise a safety component secured relative to the harness **30** and configured for coupling to an attachment element **22** of the fall protection device **20**, such as, for example a D-ring, a carabiner, and/or any other applicable fastening means attached to a distal end of the webbing **21** of the fall protection device **20**. As a non-limiting example, the mounting device **100** may be configured to detect a safe operating condition based at least in part on a determination that the fall protection device **20** defines a connected condition defined by a connected configuration of the fall protection device **20** (e.g., the attachment element **22**) relative to the operator attachment **31**, as described herein.

(19) In various embodiments, one or more of the first and second sensing devices **130**, **150** may be electronically connected to the controller **140** of the mounting device **100** by circuitry **101**. For example, the controller **140** may be electrically connected to a power source and/or one or more internal circuitries of the materials handling vehicle, so as to enable distribution of power through the controller **140** to the components of the mounting device **100** in communication therewith, such as, for example, one or more of the first and second sensing devices **130**, **150** and/or the connection condition indicator **160**. Alternatively, or additionally, the one or more of the first and second sensing devices **130**, **150** may be electrically connected to the power circuitry **101** of the materials handling vehicle **10** such that the sensing devices **130**, **150** may be powered by a power signal supplied by the power supply of the materials handling vehicle **10**. As described herein, in various embodiments, the first sensing device **130** may be configured to transmit at least a portion of the imaging data captured by the first sensing device **130** to the controller **140** via the electronic communication defined therebetween. Further, as described herein, in various embodiments, the second sensing device **150** may be configured to transmit at least a portion of the device movement data captured by the second sensing device **150** to the controller **140** via the electronic communication defined therebetween.

(20) In various embodiments, an exemplary controller **140** may be in electronically connected to the first sensing device **130** and/or a second sensing device **150** of the mounting device **100** such that the controller **140** is configured to receive captured sensor data from the first and/or second sensing device **130**, **150** and, based at least on part on the captured sensor data, detect a connection condition associated with the fall protection device **20** relative to an operator attachment **31** defined by the wearable harness **30** being worn by an operator **1**. For example, the controller **140** may be in electronically connected to the first sensing device **130** (e.g., an imaging device) such that the controller **140** may receive the captured imaging data from the imaging device **130** and, based at least on part on the captured imaging data, detect a movement condition and/or a stationary condition defined by the webbing **21** of the fall protection device **20**. As a further example, the controller **140** may be in electronically connected to the second sensing device **150** (e.g., a motion-sensing device) such that the controller **140** may receive the captured device movement data from

the second sensing device **150** and, based at least on part on the captured device movement data, detect a movement condition defined by the mounting device **100** (e.g., defined by the mounting bracket **110** and/or the rotary arm **120**). As described in further detail herein, the controller **140** may be configured to detect a connection condition associated with the fall protection device **20**, such as, for example, a connected configuration defined by a connection of an attachment element **22** to an operator attachment (e.g., defined by a wearable harness **30**), and a disconnected configuration defined by the attachment element **22** not being secured relative to the operator attachment (e.g., defined by a wearable harness **30**), based at least in part on the sensor data (e.g., imaging data, device movement data) received from first and second sensing devices **130**, **150** of the mounting device **100**.

(21) Further, in various embodiments, the mounting device **100** may further comprise a connection condition indicator **160** configured to receive one or more signals corresponding to the detected connection condition from the controller **140** and transmit at least one indication thereof, such as, for example, an audio indication signal, a visual indication signal, and/or any other appropriate signal that may be transmitted by the connection condition indicator **160** and perceived by the operator **1** and/or another party (e.g., a nearby worker, a manager, a safety coworker). In various embodiments, a connection condition indicator **160** may comprise one or more LEDs configured to be selectively operated based at least in part on an indicator signal received by the connection condition indicator **160** from the controller **140**. For example, the connection condition indicator **160** may comprise a plurality of LEDs, each being configured to selectively powered to cause an alert signal comprising a visual signal (e.g., an emitted light) that that corresponds to a particular connection condition to be transmitted from the connection condition indicator **160**. Further, in various embodiments, the connection condition indicator **160** may be configured to transmit an alert signal comprising a perceivable audio signal (e.g., via an integrated a speaker component and/or the like) that is configured to emit a predetermined sound, instructional message, and/or the like, or any combination thereof that corresponds to the connection condition detected by the controller **140**.

(22) FIGS. **2**A and **2**B illustrate various views of an exemplary mounting device for a personal fall limiter according to various embodiments described herein. In particular, FIG. **2**A and FIG. **2**B illustrate an exploded view and a side-cross-sectional view, respectively, of an exemplary mounting device **100** configured for engagement with a fall protection device **20** according to an exemplary embodiment described herein. As illustrated, in various embodiments, an exemplary mounting device **100** may comprise a mounting bracket **110** at which the mounting device **100** may be attached to a materials handling vehicle, and a rotary arm **120** rotatably connected to the mounting bracket **110** and configured for engagement with a fall protection device **20** and rotation relative to the mounting bracket **110** (e.g., about an axis of rotation **120***a*) to facilitate an expanded range of motion within the workspace for an operator operatively connected to the fall protection device (e.g., via a webbing engaged with the rotary arm **120**).

(23) In various embodiments, a mounting bracket **110** of the exemplary mounting device **100** may be configured for attachment to an upper frame element (e.g., a ceiling) of a materials handling vehicle to secure the mounting device **100** relative to the materials handling vehicle in a position at least substantially above a workspace defined by the materials handling vehicle. In various embodiments, the mounting bracket **110** may comprise an arm interface portion **111** to which the rotary arm **120** may be rotatably connected, as described herein, and one or more fastening means configured to engage at least an upper frame element of the materials handling vehicle to fixedly secure the arm interface portion **111** to the upper frame element. In various embodiments, the one or more fastening means of the mounting bracket **110** may embody any applicable fastening means configured to facilitate a coupling of the mounting bracket **110** (e.g., the arm interface portion **111**) in a fixed position above the workspace defined by the materials handling vehicle.

(24) In various embodiments, the rotary arm **120** of an exemplary mounting device **100** may be

rotatably connected to the mounting bracket **110** (e.g., at a bottom surface **111***a* of an arm interface portion **111**) such that the rotary arm **120** defines a position below the mounting bracket **110** (e.g., relative to a vertical direction defined by a ground surface upon which an exemplary materials handling vehicle to which the mounting device is coupled is positioned). In various embodiments, an attachment end (e.g., an arm base **121**) of the rotary arm **120** is secured relative to the mounting bracket **110** at via a fastening means that allows for the rotary arm **120** to rotate relative to the mounting bracket **110** about an axis of rotation **120***a* defined at the attachment end of the rotary arm **120**. For example, in various embodiments, the attachment end of the rotary arm **120** may be defined by an arm base **121** that defines a central axis extending therethrough and is connected to the mounting bracket **110** via a rotatable joint, fastener, bearing assembly, and/or any other rotatable attachment means that enables the arm base **121** to be rotated relative to the mounting bracket **110** about the central axis thereof. In various embodiments, as illustrated, the arm base **121** may be configured for attachment relative to the arm interface portion **111** of the mounting bracket **110** such that the central axis of the arm base **121** extends through the both the arm base and the mounting bracket **110** (e.g., in a direction perpendicular to the portion of the bottom surface **111***a* of the arm interface portion **111** to which that arm base **121** is operatively connected. In such an exemplary embodiment, the axis of rotation of the rotary arm **120** may be defined by central axis of the arm base **121**. As an illustrative example, in various embodiments, the mounting device **100** may be configured such that, upon the mounting bracket **110** being secured relative to an upper frame element of a materials handling vehicle, the axis of rotation **120***a* defined by the mounting device **100** may embody an at least substantially vertical axis, such as, for example, an axis in the y-direction as defined in the exemplary orientation illustrated in FIGS. **2**A and **2**B. For example, in various embodiments, an exemplary mounting device **100** may be configured such that the rotary arm **120** is configured for rotation about an axis of rotation **120***a* defined in a perpendicular direction relative to the arm interface portion **111** (e.g., a bottom surface **111***a* thereof) of the mounting bracket **110**. In such an exemplary configuration, the rotary arm **120** may be configured to rotate throughout a range of rotational motion that is defined within a rotational plane that is at least substantially parallel to the bottom surface **111***a* of the mounting bracket **110** (e.g., in a z-x plane as defined in the exemplary orientation illustrated in FIG. **2**A).

(25) In various embodiments, the arm base **121** of an exemplary rotary arm **120** may define an upper arm base portion **121***a* configured to facilitate the attachment of the arm base **121** (e.g., the rotary arm **120**) relative to the arm interface portion **111** (e.g., the bottom surface **111***a*) of the mounting bracket **110**. Further, in various embodiments, the arm base **121** of the exemplary rotary arm **120** may define a lower arm base portion **121***b* configured for receiving a fall protection device **20** to secure the fall protection device **20** relative to the rotary arm **120** and facilitate the coupling of the fall protection device **20** to the mounting device **100**. For example, at least a portion of an exemplary fall protection device **20**, such as, for example, a body portion **20***a* of the fall protection device **20**, may be attached to the lower arm base portion **121***b* via a fastening means configured for coupling the body **20***a* of the fall protection device **20** to the arm base **121**, such as, for example a hook assembly, a latch element, a snap feature, one or more guide track features, and/or any other applicable fastening means configured to facilitate the rigid attachment of the body **20***a* of the fall protection device **20** to the arm base **121** of the rotary arm **120**. For example, in various embodiments, as described herein, the fall protection device **20** may be secured to the arm base **121** of the rotary arm **120** such that the fall protection device **20** rotates with the arm base **121** through a range of rotational motion defined relative to the axis of rotation **120***a*. As such, the alignment of a webbing outlet defined by the body **20***a* of the fall protection device **20** relative to an extension arm **122** of the rotary arm **120**, as described herein, may be maintained throughout a full 360-degree rotation of the rotary arm **120** about the axis of rotation **120***a*.

(26) Further, as illustrated, the rotary arm **120** of an exemplary mounting device **100** may comprise an extension arm **122** comprising a rigid linear element (e.g., a rod and/or the like) that is rigidly

secured relative to the arm base **121** such that extension arm **122** rotates about the central axis of the arm base **121** (e.g., about the axis of rotation **120***a*) with the arm base **121**. As illustrated in FIG. **2**A, the extension arm **122** may have an arm length defined between a proximal arm end **122***a* defined at the arm base **121** and an opposing distal arm end **122***b*. In various embodiments, the extension arm **122** may extend from a portion of the arm base **121** in a radially outward direction relative to central axis of the arm base **121** such that the distal arm end **122***b* of the extension arm **122** defines a radially extended position separated from the arm base **121** (e.g., the central axis thereof) by an extension distance (e.g., as measured in a radially outward direction relative to the axis of rotation **120***a*) that is defined at least in part by the arm length of the extension arm **122**. For example, the rotary arm **120** may be configured such that a rotation of the rotary arm **120** throughout the range of rotational motion thereof (e.g., about the axis of rotation **120***a*) may include the distal arm end **122***b* being rotated within a rotational plane that is at least substantially perpendicular to the axis of rotation **120***a* defined by the rotary arm **120** (e.g., in a z-x plane as defined in the exemplary orientation illustrated in FIG. **2**A). As a non-limiting illustrative example, the rotary arm **120** of the exemplary mounting device **100** may be configured such that the rotational plane within which the distal arm end **122***b* moves as the rotary arm **120** is rotated throughout the range of rotational motion (about the axis of rotation **120***a*) may embody an at least substantially horizontal plane (e.g., relative to a horizontal direction defined by a ground surface upon which an exemplary materials handling vehicle to which the mounting device is coupled is positioned).

(27) In various embodiments, the extension arm **122** of the rotary arm **120** may define a webbing retention feature **123** configured for receiving a portion of a webbing **21** of a fall protection element **20** defined between the body **20***a* and an attachment element **22** to define an intermediate webbing anchor position that is moveable with the rotation of the rotary arm **120** to facilitate an expanded range of motion within a workspace for an operator that is operatively attached to the fall protection device **20** (e.g., via the webbing **21**). In various embodiments, as illustrated, the webbing retention feature **123** may be defined along the extension arm **122** in a position at least substantially proximate the distal arm end **122**. The webbing retention feature **123** may be configured to facilitate the arrangement of at least a portion of the webbing **21** (e.g., an intermediate webbing portion thereof) relative to the distal arm end **122***b* of the extension arm **122** by enabling the webbing **21** to be threaded through the webbing retention feature **123** such that the webbing **21** remains in contact with the extension arm **122** throughout the rotation of the rotary arm **120** through the range of rotational motion, as described herein.

(28) For example, in various embodiments, an exemplary mounting device **100**, as illustrated in FIGS. **2**A and **2**B, may be configured to receive the webbing **21** within the webbing retention feature **123** by the webbing **21** being provided from the body **20***a* of the fall protection device **20** mounted to the arm base **121** through the webbing retention feature **123** in an outward direction (e.g., away from the central axis of the arm base **121**). The extension arm **122** may be configured such that, upon the webbing **21** being provided through the webbing retention feature **123**, the webbing **21** may physically contact (e.g., wrap around) at least a portion of the extension arm **122** defined at the distal arm end **122***b* such that the remaining portion of the webbing **21** defined between the webbing retention feature **123** and the attachment element **22** is suspended from the distal arm end **122***b* in an at least partially downward vertical direction (e.g., in the negative y-direction according to the exemplary orientations illustrated in FIGS. **2**A and **2**B). As shown, the webbing retention feature **123** is configured to receive the webbing **21** therethrough such that, rather than the webbing **21** extending directly from the body **20***a* of the fall protection device **20** to the attachment element **22**, the webbing **21** is guided through the webbing retention feature **123** as it extends along a webbing length thereof from body **20***a* to the attachment element **22**. In various embodiments, the mounting device **100** may be engaged with the fall protection device **20** such that a first length portion **21***a* of the webbing **21** extends from the body **20***a* in a radially outward

direction (e.g., relative to the axis of rotation **120***a*) to the webbing retention feature **123**, and a second length portion **21***b* of the webbing **21** extends from the webbing retention feature **123** in a second direction defined in an at least partially vertical direction to the attachment element **22** that may be operatively secured relative to an operator positioned within a workspace defined below the mounting device **100**. For example, the second length portion **21***b* of the webbing **21** provided between the distal arm end **122***b* (e.g., the webbing retention feature **123**) and an operator attachment (e.g., a wearable harness) may define a subrange of operator motion that is defined by an area within the workspace throughout which an operator may move relative to the distal arm end **122***b* without causing the rotary arm **120** to be rotated (e.g., as a result of drag and/or tension forces caused by a an operator movement pulling on the webbing **21**).

(29) In various embodiments wherein the attachment element **22** of the fall protection device **20** is connected to an operator attachment, an exemplary mounting device **100** may be configured such that one or more forces generated by a movement of the operator within the workspace (e.g., an operator movement imparting pulling force on the operator attachment) may be transmitted from the operator attachment to the rotary arm **120** via the webbing **21** (e.g., the second length portion **21***b*). For example, the one or more forces may act on the rotary arm **120** at the webbing retention feature **123** defined by the extension arm **122** (e.g., an extended webbing engagement point) such that a non-linear torque and a moment are imparted on the extension arm **122**, causing the rotary arm **120** to rotate in a corresponding rotational direction about the axis of rotation **120***a*. As described herein, the arrangement of the webbing within the webbing retention feature **123** and the physical engagement of the webbing **21** with the extension arm **122** may enable at least a portion of the one or more forces present within the webbing **21** (e.g., from the operator attachment) as a result of a movement of the operator throughout the workspace to be at least partially mitigated by the mounting device **100** prior to being transmitted to fall protection device **20**. In various embodiments, the mounting device **100** being configured to receive an intermediate length portion of the webbing **21** of the fall protection device **20** at a webbing retention feature **123** defined by a rotary arm **120** that is rotatable about an axis of rotation **120***a* enables an operator that operatively connected to the fall protection device **20** to experience an at least substantially reduced (e.g., minimized) drag force from the fall protection device **20** as the operator moves throughout the workspace.

(30) The portion of the extension arm **122** at which the webbing **21** physically contacts the extension arm **122** (e.g., within the webbing retention feature **123** and/or upon the webbing being provided therethrough) may define an extended webbing engagement point from which the webbing **21** may extend from the rotary arm **120** to the attachment element **22**. The extension arm **122** may be configured such that the extended webbing engagement point defined at the distal arm end **122***b* (at least substantially adjacent the webbing retention feature **123**) is spatially separated from the arm base **121** and the body **20***a* of the fall protection device **20** secured thereto by a radial distance measured relative to the axis of rotation **120***a*. For example, the extended webbing engagement point may correspond at least in part to the radially extended position of the distal arm end **122***b* of the rotary arm **120** relative to the axis of rotation **120***a*. In such an exemplary configuration, as the extension arm **122** is rotated throughout a range of rotational motion, the extended webbing engagement point is similarly rotated about the axis of rotation **120***a* such that a subrange of operator motion defined relative to the extended webbing engagement point (e.g., at the webbing retention feature **123** of the rotary arm **120**) is relocated (e.g., shifted) from a first area within the workspace to a second area within the workspace. An exemplary mounting device **100** may be configured to enable the rotary arm **120** to be rotated 360 degrees about the axis of rotation **120***a* defined by the central axis of the arm base **121** in both clockwise and counter-clockwise rotational directions, depending on the movement(s) of an operator through the workspace defined by the materials handling vehicle. For example, as the rotary arm **120** is rotated through a full 360-degree range of rotational motion, the collective area covered by the shifting subranges of operator

motion defined relative to the rotating extended webbing engagement point including both the first and second areas described above—may define an expanded overall range of motion within the workspace.

(31) The rotary arm **120** of the mounting device **100** being rotatably connected to the mounting bracket **110** and defining a 360-degree range of rotational motion reduces the amount of drag force that is realized by the webbing **21** as an operator operatively connected to an attachment element **22** of the fall protection device **20** moves throughout the workspace below the mounting device **100**. Further, the mounting device **100** being configured to reduce the drag forces associated with operator movement about the workspace enables a fall protection device **20** secured thereto to be connected to the operator (e.g., via an attachment element **22**) with a minimal amount of slack (e.g., excess material length) present in the webbing **21** during operation. For example, traditional means of providing an excess length of webbing **21** between the body **20***a* and the attachment element **22** to reduce the drag force realized by the fall protection device **20** during operator movement are overcome by the exemplary mounting device **100** described herein, enabling a minimization of the length of the webbing **21** that corresponds to a safer operating condition for the operator.

(32) In various embodiments, an exemplary mounting device **100** may further comprise a first sensing device configured to detect a position and/or movement of at least a portion of a webbing **21** provided within the webbing retention feature **123** relative to the **123**. For example, in various embodiments, the first sensing device may comprise an imaging device **130** configured to capture imaging data comprising at least one image that shows at least a portion of the webbing **21** of a fall protection device **20** that is provided within, at, and/or at least substantially adjacent to the webbing retention feature **123** defined by the rotary arm **120** of the mounting device **100**. In various embodiments, the imaging device **130** may at least substantially continuously, serially, and/or periodically capture image data including a plurality of images and/or the like that may be at least substantially continuously processed (e.g., by the controller **140**) and/or analyzed such that the exemplary mounting device **100** is configured to at least substantially continuously monitor the position and/or movement(s) of the webbing **21** within the webbing retention feature **123** (e.g., relative to the imaging device **130**) that may be utilized to determine an connection condition of the fall protection device **20** relative to an operator attachment at a plurality of instances in series. As described herein, the imaging device **130** may embody an optical sensor secured to the extension arm **122** and having a line of sight defined directly between the imaging device **130** and the webbing retention feature **123**.

(33) In various embodiments, the imaging device **130** may be fixedly secured relative to the rotary arm **120** of the mounting device **100**. For example, FIG. **3** illustrates an exemplary mounting device **100** comprising an imaging device **130** that is secured to the rotary arm **120** at a position along the extension arm **122** that is configured for movement with the extension arm **122** (e.g., the distal arm end **122***b*) such that the imaging device **130** does not move relative to the webbing retention feature **123**. For example, in various embodiments, the imaging device **130** may be secured to the rotary arm **120** in a position relative to the extension arm **122** such that the imaging device **130** has a direct line of sight to the webbing retention feature **123** defined by the extension arm **122**, such as, for example, at a position within the webbing retention feature **123**, at the distal arm end **122***b* of the extension arm **122**, and/or otherwise at least substantially proximate webbing retention feature **123**.

(34) In various embodiments, the imaging device **130** may comprise an imaging component, such as, for example, a camera having a resolution of at least 720p and being configured to capture imaging data comprising videos, images, and/or the like, an optical lens configured to define a field of view of the imaging device **130**, an imaging device processing unit, various internal circuitries configured to the facilitate power management and connectivity and/or networked communication of the imaging device **130**. In various embodiments, the imaging device **130** may have a designated field of view for capturing, permanently and/or temporarily, one or more images of at least a

portion of the webbing **21** of a fall protection device **20** that is provided within the webbing retention feature **123** of the mounting device **100**. For example, in various embodiments, the imaging device **130** may be positioned relative to the webbing retention feature **123** of the rotary arm **120** such that at least a portion of a width of the webbing **21** provided within the webbing retention element **123** is disposed within the field of view of the imaging device **130**. As a non-limiting example, in various embodiments, the imaging device **130** may be positioned relative to the webbing retention element **123** such that the optical lens of the imaging device **130** is separated from the portion of a width of the webbing **21** provided within the webbing retention element **123** by a distance of at least approximately between 10.0 mm and 25.0 mm. As a further example, in various embodiments, based at least in part on the positioning of the imaging device **130** (e.g., relative to the webbing retention feature **123**), the optical lens of the imaging device may be configured such that the field of view of the imaging device **130** may be at least approximately between 1 degree and 180 degrees (e.g., between 15 degrees and 45 degrees. In various embodiments, the imaging device processing unit of the imaging device **130** may comprise one or more hardware components and/or circuitries that are distinct from the controller **140** of the mounting device **100**. Alternatively, or additionally, in various embodiments, one or more hardware components, circuitries, and/or functionalities of the imaging device processing unit of the imaging device **130** may be defined by the controller **140**, as described herein, such that the imaging device processing unit of the imaging device **130** may be defined as part of the controller **140**.

(35) In various embodiments, the imaging device **130** may be configured to capture imaging data of the webbing **21** provided within the webbing retention feature **123** to facilitate a detection of a movement, such as, for example, a micro-movement, of the webbing **21** relative to the extension arm **122** (e.g., within the webbing retention feature **123**). As a non-limiting example provided for illustrative purposes, in various embodiments, the imaging device **130** of an exemplary mounting device **100** may be positioned within the webbing retention feature **123** and may comprise a sensing means configured to capture imaging data to execute a motion-sensing operation that is at least substantially similar to the motion-sensing operation executed by a computer mouse to detect a movement thereof relative to an underlying surface. That is, in various embodiments, the imaging device **130** may be positioned within the webbing retention feature **123** and configured to capture imaging data to detect and/or characterize a relative movement, and/or lack thereof, of the portion of the webbing **21** disposed within the webbing retention feature **123** with respect to one or more adjacent surfaces of the extension arm **122** that define the webbing retention feature **123**. The imaging data captured by the imaging device **130** may be utilized by the mounting device **100** to detect one or both of a movement condition defined by a detected movement (e.g., a micro-movement) of the webbing **21** portion within the webbing retention feature **123** relative to the imaging device **130** and/or an adjacent surface of the extension arm **122**, and a stationary condition defined by the webbing **21** portion within the webbing retention feature **123** exhibiting at least substantially negligible movement relative to the imaging device **130** and/or the adjacent surface of the extension arm **122**. As described herein, the detection of a movement condition and/or a stationary condition by the imaging device **130**, as defined by the webbing **21** within the webbing retention feature **123**, may define an input variable that may be utilized by the mounting device **100** in combination with one or more other input variables defined by mounting device sensor data, such as, for example, device movement data captured by a second sensing device that detects and/or characterizes any movement of the mounting bracket **110** and/or the rotary arm **120**, to determine whether the attachment element **22** of the fall protection device **20** is attached to the operator attachment **31**.

(36) Further, in various embodiments, an exemplary mounting device **100** may further comprise a second sensing device configured to detect a movement of the mounting device **100** secured relative to the materials handling vehicle (e.g., a linear movement of the mounting bracket **110**, a rotational movement of the rotary arm **120**, and/or the like). In various embodiments, as further

illustrated in FIG. **3**, the second sensing device may comprise a motion-sensing device **150** configured to capture sensor data that correspond to one or more movements, or lack thereof of, of the mounting device **100** in any and/or all of six directions (e.g., defining six degrees of freedom). For example, the motion-sensing device **150** may comprise a motion-sensing device configured to execute a six-degrees-of-freedom ("6DOF") motion-sensing operation in order to detect and/or characterize any movement of the mounting bracket **100**. In various embodiments, the motion-sensing device **150** may comprise a gyroscope and an accelerometer that are each configured to detect movement of the mounting bracket **100** in a respective direction based on one or more inertial measurements captured as device movement data by the motion-sensing device **150**. The motion-sensing device **150** may at least substantially continuously, serially, and/or periodically capture device movement data that may be at least substantially continuously processed (e.g., by the controller **140**) and/or analyzed such that the exemplary mounting device **100** is configured to at least substantially continuously monitor the movements of the mounting device **100** at a plurality of instances in series.

(37) As illustrated in the exemplary mounting device **100** shown in FIG. **3**, in various embodiments, the motion-sensing device **150** may be fixedly secured relative to the mounting bracket **110** of the mounting device **100**. As illustrated, the motion-sensing device **150** may be secured to the mounting bracket **110** such that a movement of the materials handling vehicle **10** in one or more directions during an operation thereof may correspond to a similar movement of the mounting bracket **110** rigidly secured thereto (e.g., at the upper frame element **12**), and thus, may define a movement of the motion-sensing device **150** secured to the mounting bracket **110** that may be detected and/or characterized by the motion-sensing device **150**. Further, the motion-sensing device **150** may be secured to the mounting bracket **110** at a position that enables the motion-sensing device **150** to detect a movement (e.g., a rotation) of the rotary arm **120** relative to the mounting bracket **110**.

(38) In various embodiments, the detection of a movement of the mounting bracket **100** by the motion-sensing device **150** may define an input variable that may be utilized by the mounting device **100** in combination with a second input variable defined the movement condition and/or a stationary condition defined by the webbing **21** within the webbing retention feature **123**, as captured by the imaging device **130**, to determine whether the attachment element **22** of the fall protection device **20** is attached to the operator attachment **31**. For example, the motion-sensing device **150** may be configured to capture device movement data associated with the mounting device **100** (e.g., the mounting bracket **110** and/or the rotary arm **120**) that corresponds to one or more movements of the materials handling vehicle **10** and/or the operator **1** positioned within the workspace **11**. Accordingly, the device movement data captured by the motion-sensing device **150** may be utilized by the mounting device **100** to determine whether the webbing **21** within the webbing retention feature **123** should define movement condition and/or the stationary condition at a particular instance based at least in part on whether a movement of the mounting bracket **100** was detected by the motion-sensing device **150** at that particular instance. As described herein, the mounting device **100** may be configured to determine that the attachment element **22** of the fall protection device **20** defines a disconnected condition relative to the operator attachment **31** based on a detection that the webbing **21** within the webbing retention feature **123** defines a stationary condition (e.g., as captured by the imaging device **130**) during a detected movement of the mounting device **100** (e.g., as captured by the motion-sensing device **150**).

(39) As described herein, in various embodiments, the mounting device **100** may comprise a first sensing device and a second sensing device configured to at least substantially continuously, serially, and/or periodically capture data (e.g., imaging data, device movement data, and/or the like) that may be at least substantially continuously processed (e.g., by the controller **140**) to at least substantially continuously monitor and/or determine a plurality of data outputs, such as, for example, a movement condition and/or a stationary condition defined by the webbing **21** within the

webbing retention feature **123** and a detected movement condition defined by the mounting device **100** (e.g., a linear movement of the mounting bracket **110** and/or a rotation of the rotary arm **120**), which may utilized to collectively determine and/or define an installation condition of the fall protection device **20** relative to an operator attachment at a particular instance (e.g., instantaneously) and/or at a plurality of sequential instances (e.g., serially).

(40) Further, in various embodiments, the rotary arm **120** (e.g., the arm base **121**) may define a hollow interior through which one or more wires (e.g., power circuitry) may be provided in order to enable the electronic communication of the first sensing means **130** with one or more other electronic components of the mounting device **100** and/or the materials handling vehicle, such as, for example, a controller of the mounting device, the power circuitry of the materials handling vehicle, and/or the like. As described herein, the one or more wires may extend through the arm base **121** of the rotary arm **120** and be directed along the arm length of the extension arm **122** to facilitate an electronic connection with the first sensing means **130** (e.g., at the webbing retention feature **123**). In various embodiments, the fastening means used for securing the rotary arm **120** relative to the mounting bracket **110** and defining the rotatable configuration of the rotary arm **120** may include various fastening elements such as bearings, discs, rings, and/or the like that enable the arm base **121** of the rotary arm **120** to be rotated through a full 360-degree range of rotational motion without causing the one or more wires disposed therethrough to be twisted and/or otherwise undesirably rearranged. For example, the rotary arm **120** may utilize one or more slip rings configured to allow various power and/or control circuitries provided within the arm base **121** to remain untwisted and/or avoid further undesirable twisting as the arm base **121** of the rotary arm **120** is rotated through a 360-degree range of rotation, as described herein.

(41) As illustrated in FIG. **4**, an exemplary mounting device **100** may comprise a controller **140** comprising a memory **141**, a processor **142**, input/output circuitry **143**, communication circuitry **144**, an imaging device data repository **107**, materials handling vehicle control circuitry **145**, image processing circuitry **146**, fall protection device connection condition detection circuitry **147**, and alert management circuitry **148**. The controller **140** may be configured to execute the operations described herein. Although the components are described with respect to functional limitations, it should be understood that the particular implementations necessarily include the use of particular hardware. It should also be understood that certain of the components described herein may include similar or common hardware. For example, two sets of circuitry may both leverage use of the same processor, network interface, storage medium, or the like to perform their associated functions, such that duplicate hardware is not required for each set of circuitry.

(42) The term “circuitry” should be understood broadly to include hardware and, in some embodiments, software for configuring the hardware. For example, in some embodiments, “circuitry” may include processing circuitry, storage media, network interfaces, input/output devices, and the like. In some embodiments, other elements of the controller **140** may provide or supplement the functionality of particular circuitry. For example, the processor **142** may provide processing functionality, the memory **141** may provide storage functionality, the communication circuitry **144** may provide network interface functionality, and the like.

(43) In some embodiments, the processor **142** (and/or co-processor or any other processing circuitry assisting or otherwise associated with the processor) may be in communication with the memory **141** via a bus for passing information among components of the device. The memory **141** may be non-transitory and may include, for example, one or more volatile and/or non-volatile memories. For example, the memory **141** may be an electronic storage device (e.g., a computer readable storage medium). In various embodiments, the memory **141** may be configured to store information, data, content, applications, instructions, or the like, for enabling the device to carry out various functions in accordance with example embodiments of the present disclosure. It will be understood that the memory **141** may be configured to store partially or wholly any electronic information, data, data structures, embodiments, examples, figures, processes, operations,

techniques, algorithms, instructions, systems, apparatuses, methods, look-up tables, or computer program products described herein, or any combination thereof. As a non-limiting example, the memory **141** may be configured to store data captured by a first sensing device of the mounting device **100** (e.g., imaging data captured by an imaging device), corresponding data generated by the controller **140** of the mounting device **100**, timestamp data, location data, historical data and/or the like, associated with a workspace (e.g., a materials handling vehicle).

(44) The processor **142** may be embodied in a number of different ways and may, for example, include one or more processing devices configured to perform independently. Additionally or alternatively, the processor may include one or more processors configured in tandem via a bus to enable independent execution of instructions, pipelining, and/or multithreading. The use of the term “processing circuitry” may be understood to include a single core processor, a multi-core processor, multiple processors internal to the apparatus, and/or remote or “cloud” processors.

(45) In an example embodiment, the processor **142** may be configured to execute instructions stored in the memory **141** or otherwise accessible to the processor. Alternatively, or additionally, the processor may be configured to execute hard-coded functionality. As such, whether configured by hardware or software methods, or by a combination thereof, the processor may represent an entity (e.g., physically embodied in circuitry) capable of performing operations according to an embodiment of the present disclosure while configured accordingly. Alternatively, as another example, when the processor is embodied as an executor of software instructions, the instructions may specifically configure the processor to perform the algorithms and/or operations described herein when the instructions are executed.

(46) In some embodiments, the controller **140** may include input-output circuitry **143** that may, in turn, be in communication with the processor **142** to provide output to a user and, in some embodiments, to receive input such as a command provided by the user. The input-output circuitry **143** may comprise a user interface, such as a graphical user interface (GUI), and may include a display that may include a web user interface, a GUI application, a mobile application, a client device, or any other suitable hardware or software. In some embodiments, the input-output circuitry **143** may also include a display device, a display screen, user input elements, such as a touch screen, touch areas, soft keys, a keyboard, a mouse, a microphone, a speaker (e.g., a buzzer), a light emitting device (e.g., a red light emitting diode (LED), a green LED, a blue LED, a white LED, an infrared (IR) LED, or a combination thereof), or other input-output mechanisms. The processor **142**, input-output circuitry **143** (which may utilize the processing circuitry), or both may be configured to control one or more functions of one or more user interface elements through computer-executable program code instructions (e.g., software, firmware) stored in a non-transitory computer-readable storage medium (e.g., memory **141**). Input-output circuitry **143** is optional and, in some embodiments, the controller **140** may not include input-output circuitry. For example, in various embodiments, the controller **140** may generate one or more alert signals (e.g., data) to be transmitted to one or more other devices with which one or more authorized users (a manager, safety coordinator, and/or the like) directly interact and cause the one or more alert signals to be transmitted at the one or more other devices.

(47) The communication circuitry **144** may be a device or circuitry embodied in either hardware or a combination of hardware and software that is configured to receive and/or transmit data from/to a network and/or any other device, circuitry, or module in communication with the controller **140**. For example, the communication circuitry **144** may be configured to communicate with one or more computing devices via wired (e.g., USB) or wireless (e.g., Bluetooth, Wi-Fi, cellular, and/or the like) communication protocols. For example, in various embodiments, the communication circuitry **144** may be configured to facilitate data communication between an exemplary mounting device **100** and one or more external computing devices via wired (e.g., USB, ethernet, and/or the like) and/or wireless (e.g., Bluetooth, Wi-Fi, cellular, and/or the like) communication protocols.

(48) In various embodiments, the processor **142** may be configured to communicate with the

materials handling vehicle control circuitry **145**. The materials handling vehicle control circuitry **145** may be a device or circuitry embodied in either hardware or a combination of hardware and software that is configured to facilitate an operation of a materials handling vehicle **10** by generating a control signal configured to operate one or both of a drive assembly and a lift assembly of the materials handling vehicle **10** in response to an interaction of an operator **1** with one or more vehicle operation controls (e.g., user controls) of the materials handling vehicle **10** from an operating position within a workspace **11**. In various embodiments, the materials handling vehicle control circuitry **145** may be configured to receive a first control signal based on operator interaction with the vehicle operation controls of the materials handling vehicle **10**), and, in response, transmit a corresponding signal to one or more circuitries of the controller **140**, such as, for example, the fall protection device connection condition detection circuitry **147**, in order to facilitate detection of a connection condition (e.g., a connected configuration) based at least in part on the materials handling vehicle **10** being operated by the operator at an instance when the controller **140** determines (e.g., based on imaging data from a first sending device) that the webbing of a fall protection device at a webbing retention feature defines a movement condition. As a non-limiting example provided for illustrative purposes, in an exemplary circumstance wherein the materials handling vehicle control circuitry **145** receives a first control signal embodying a user operation of the drive assembly and/or the lift assembly of the materials handling vehicle **10**, the materials handling vehicle control circuitry **145** may be configured to transmit a corresponding signal to the fall protection device connection condition detection circuitry **147** so as to enable a detection of a connection condition defined by the fall protection device **20** relative to an operator attachment, based at least in part on a movement condition and/or a stationary condition defined by the webbing of the fall protection device at a webbing retention feature of a mounting device **100**, as determined by imaging data captured by a first sensing device **130** (e.g., an imaging device).

(49) For example, in an exemplary embodiment wherein the materials handling vehicle control circuitry **145** receives a first control signal embodying a user operation of the drive assembly and/or the lift assembly of the materials handling vehicle **10**, and the fall protection device connection condition detection circuitry **147** detects a movement condition defined by the webbing of the fall protection device at the webbing retention feature of the mounting device **100**, the mounting device **100** may be configured to determine that the attachment element of the fall protection device defines a connected configuration relative to the operator attachment. In such an exemplary configuration, the controller **140** (e.g., the fall protection device connection condition detection circuitry **147**) may detect a connection condition associated with the fall protection device based at least in part on the connected configuration detected by the controller **140**. Further, in an exemplary embodiment wherein the materials handling vehicle control circuitry **145** receives a first control signal embodying a user operation of the drive assembly and/or the lift assembly of the materials handling vehicle **10**, and the fall protection device connection condition detection circuitry **147** detects a stationary condition defined by the webbing of the fall protection device at the webbing retention feature of the mounting device **100**, the mounting device **100** may be configured to determine that the attachment element of the fall protection device defines a disconnected configuration relative to the operator attachment. In such an exemplary configuration, the controller **140** (e.g., the fall protection device connection condition detection circuitry **147**) may detect an uninstalled condition associated with the fall protection device based at least in part on the disconnected configuration detected by the controller **140**.

(50) In various embodiments, at least a portion of the controller **140**, such as, for example, at least a portion of the materials handling vehicle control circuitry **145** and/or the processor **142** may be at least substantially integrated with the circuitry of the materials handling vehicle itself. For example, in various embodiments, one or more signals generated by the controller **140** (e.g., from the materials handling vehicle control circuitry **145**, the communications circuitry **144**, and/or the

processor **142**) in association with a connection condition may comprise a lead signal that may be received by the materials handling vehicle circuitry (e.g., controls) and configured to cause the materials handling vehicle **10** to be controlled and/or operated in accordance with the lead signal. As a further example, in various embodiments, one or more signals generated by the controller **140** (e.g., from the materials handling vehicle control circuitry **145**, the communications circuitry **144**, and/or the processor **142**) in association with a connection condition may comprise a slave signal that may be generated in response to one or more signals received from the materials handling vehicle circuitry (e.g., controls). Additionally, or alternatively, in such an exemplary system architecture wherein the controller **140** is configured to generate a slave signal that is received by the materials handling vehicle (e.g., controls), the slave signal may comprise a passive and/or informative data signal that is not configured to directly cause a reactive operation and/or action by the by the materials handling vehicle circuitry, but rather may be configured for processing by the materials handling vehicle, which may be configured to generate a reactionary signal based at least in part on the data contained in the slave signal.
(51) In various embodiments, the processor **142** may be configured to communicate with the image processing circuitry **146**. The image processing circuitry **146** may be a device and/or circuitry embodied in either hardware or a combination of hardware and software that is configured to receive, process, generate, and/or transmit data (e.g., imaging data), such as one or more images, videos, and/or the like captured by a first sensing device (e.g., an imaging device **130** as illustrated in the exemplary mounting device **100** of FIG. **3**). In various embodiments, the image processing circuitry **146** may be further configured to analyze the one or more images captured by the imaging device using at least one processing technique to determine one or more characteristics of a connection condition as defined by a connection configuration and/or a disconnected configuration associated with the fall protection device as captured in the sensor data (e.g., imaging data, device movement data) captured by the first and second sensing elements of the mounting device **100**.
(52) In various embodiments, the image processing circuitry **146** may send and/or receive imaging data captured by the imaging device **130** and/or corresponding data associated therewith generated in a supported format by the image processing circuitry **146** to and/or from the imaging device data repository **107**.
(53) Further, in various embodiments, the image processing circuitry **146** may be configured to analyze imaging data comprising one or more images captured by the imaging device **130** of the mounting device **100** and/or a fall protection device (e.g., a webbing length portion) engaged with the device **100** to detect and/or characterize a change in one or more conditions (e.g., positions, characteristics, configurations, and/or the like) between a first time and a second time, such as, for example, a movement (e.g., a micro-movement) and/or change in position of at least a portion of a webbing **21** provided within the webbing retention feature **123** of the extension arm **122** relative to the webbing retention feature **123**, and/or the like. The image processing circuitry **146** may receive from the imaging device **130**, for example, a first captured image and a second captured image, captured at the first time and the second time, respectively, wherein the second time is subsequent the first time (occurs after the first time). In such a configuration, the image processing circuitry **146** may be configured to distinguish between a first condition as defined in the first captured image and a second condition as defined in the second captured image by comparing the respective images captured at the first and second times and identifying any positions, characteristics, configurations, and/or conditions that were at least substantially different as defined in the second captured image.
(54) In various embodiments, the processor **142** may be configured to communicate with the fall protection device connection condition detection circuitry **147**. The fall protection device connection condition detection circuitry **147** may be a device and/or circuitry embodied in either hardware or a combination of hardware and software that is configured to receive, process, generate, and/or transmit data (e.g., sensor data), such as imaging data and/or device movement

data captured by a mounting device **100** (e.g., via a first sensing device **130** and/or a second sensing device **150**) and/or data corresponding thereto in order to detect the connection condition associated with a fall protection device relative to the operator attachment.

(55) As described herein, in various embodiments, the fall protection device connection condition detection circuitry **147** may be configured to detect a connection condition associated with a fall protection device with respect to an operator attachment based at least in part on captured imaging data associated with a webbing portion of the fall protection device and captured device movement data associated with the mounting device (e.g., the mounting bracket and/or the rotary arm). In particular, as described herein, the fall protection device connection condition detection circuitry **147** may be configured to detect a connection condition defined by a connected configuration and/or a disconnected configuration of the attachment element of the fall protection device relative to the operator attachment. For example, the fall protection device connection condition detection circuitry **147** may be configured to at least facilitate a detection of a connection condition that is defined by a connected configuration of a fall protection device (e.g., an attachment element) relative to an operator attachment by determining that the fall protection device is in a connected configuration relative to the operator attachment during operation of the materials handling vehicle. The mounting device **100** may be configured to detect a connection condition defined by a connected configuration of the attachment element relative to the operator attachment based on first sensor data (e.g., imaging data captured by the first sensing device) corresponding to a movement condition defined by a movement of the webbing of a fall protection device relative to a webbing retention feature defined by the rotary arm, and second sensor data corresponding to any movement of the mounting bracket **100**, including a linear movement of the mounting bracket (e.g., within an environment, as defined by a movement of the materials handling vehicle to which the mounting device **100** is secured) and/or a rotational movement of the rotary arm about the axis of rotation.

(56) In various embodiments, the fall protection device connection condition detection circuitry **147** may be configured to at least facilitate a detection of a connection condition that is defined by a disconnected configuration of a fall protection device (e.g., an attachment element) relative to an operator attachment by determining that the fall protection device is in a disconnected configuration relative to the operator attachment during operation of the materials handling vehicle. For example, the fall protection device connection condition detection circuitry **147** may be configured to at least facilitate a detection of a disconnected configuration of the attachment element of the fall protection device relative to an operator attachment by determining that the portion of the webbing of the fall protection device disposed at the webbing retention feature of the rotary arm **120** defines a stationary condition during operation of the materials handling vehicle **10**, such as, for example, at an instance in which the controller **140** detects one or more control signals associated with operator instructions to control/operate the materials handling vehicle, or a movement condition defined by the mounting device **100** based at least in part on sensor data captured by a second sensing device

(57) In various embodiments, the fall protection device connection condition detection circuitry **147** may be configured such that, upon detecting a connection condition defined by a disconnected configuration of the fall protection device (e.g., an attachment element) relative to an operator attachment, the fall protection device connection condition detection circuitry **147** may transmit at least one signal comprising a control signal configured to cause at least a portion of the operation functionalities of the materials handling vehicle to be suspended. For example, in such an exemplary circumstance, in various embodiments, the fall protection device connection condition detection circuitry **147** may transmit at least one signal comprising a control signal configured to cause the materials handling vehicle (e.g., the drive controls, the lift controls) to be shut down, thereby preventing the materials handling vehicle from being operated while the mounting device **100** detected a dangerous condition.

(58) In various embodiments, the fall protection device connection condition detection circuitry

147 may be configured such that, upon detecting the connection configuration associated with the fall protection device relative to the operator attachment, the fall protection device connection condition detection circuitry **147** may transmit at least one signal and/or corresponding data (e.g., data indicative of the fall protection device being configured in a fully-installed configuration and/or in an uninstalled configuration) to one or more of the alert management circuitry **148**, the input/output circuitry **143**, and/or the imaging device data repository **107**, such as, for example, in order to facilitate a determination of a connection condition.
(59) In various embodiments, the processor **142** may be configured to communicate with the alert management circuitry **148**. The alert management circuitry **148** may be a device or circuitry embodied in either hardware or a combination of hardware and software that is configured to, upon a connection condition associated with the fall protection device being detected by the exemplary mounting device **100**, cause one or more alert signals corresponding to the detected connection condition to be transmitted from a connection condition indicator of mounting device **100**. As described herein, the alert management circuitry **148** may facilitate transmission (e.g., emission, display, and/or any other perceivable means of signal communication) of an exemplary alert signal comprising an audio and/or visual signal corresponding to a detected connection condition that, upon being transmitted from the connection condition indicator, may embody a perceivable indication of the connection condition (e.g., a light, a sound, a message, and/or the like, or any combination thereof) and/or an instructional message corresponding to the detected connection condition. For example, the alert management circuitry **148** may generate an indicator signal corresponding to the detected connection condition and may cause the indicator signal to be transmitted to the connection condition indicator for transmission therefrom as one or more audio and/or visual alert signals embodying an indication of the connection condition detected by the mounting device **100**.
(60) In various embodiments, an exemplary mounting device **100** may be configured with, or in communication with, an imaging device data repository **107**. The imaging device data repository **107** may be stored, at least partially on the memory **141** of the system. In some embodiments, the imaging device data repository **107** may be remote from, but in connection with, the mounting device **100**. The imaging device data repository **107** may contain information, such as images relating to one or more materials handling vehicles (e.g., workspaces), fall protection devices, fastening/attachment means, webbing types, and/or the like. In some embodiments, the mounting device **100** may also use machine learning for detecting one or more conditions, configurations, and/or the like associated with a fall protection device in order to facilitate the detection of a connection condition associated with the fall protection device, such that the mounting device **100** may use a reference database, such as the imaging device data repository **107**, to initially train the mounting device **100** and then may be configured to detect a connection condition without referencing the imaging device data repository **107** or other reference databases. For example, in various embodiments, a controller **140** may be configured to execute a feedback loop, wherein one or more imaging data, device movement data, corresponding connected configurations, and/or determined characteristics (e.g., characteristics associated with an operator, a fall protection device, and/or a materials handling vehicle) associated with the sensor data captured by the sensing devices may define one or more inputs into a machine learning model in order to increase a rate of machine learning associated with the one or more machine learning techniques, as described herein.
(61) In various embodiments, although various operations described herein with respect to the mounting device (e.g., controller **140**) may be described, illustrated, and/or otherwise disclosed for illustrative purposes as sequential operations executed in series, it should be understood that at least a portion of operations executed and/or facilitated by the mounting device **100** (e.g., controller **140**) may comprise synchronous operations and/or communications that may be executed (e.g., by an exemplary mounting device **100**) at least substantially simultaneously. For example, in various embodiments, a connection condition may be detected at an at least substantially singular instance

based on a plurality of data outputs corresponding to a plurality of configurations relating to materials handling vehicles, fall protection devices, workspaces and/or operator configurations (e.g., positions, orientations, connections, and/or the like) as captured by an exemplary imaging device in association with at least substantially the same instance.

(62) Many modifications and other embodiments will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A mounting device for a fall protection device, the mounting device comprising: a mounting bracket configured for attachment to a materials handling vehicle; and a rotary arm rotatably attached to the mounting bracket at an arm base, wherein the rotary arm is configured to rotate relative to the mounting bracket about an axis of rotation defined at the arm base, wherein the rotary arm is rotatably attached to the mounting bracket via one or more fastening elements comprising a slip ring, wherein the rotary arm defines a webbing retention feature configured for receiving a webbing of the fall protection device to define a dynamic engagement of the rotary arm with an intermediate webbing portion defined along a length of the webbing, the webbing retention feature defining an extended webbing engagement point, and wherein the webbing retention feature is defined at a distal position along an extension arm of the rotary arm, and wherein a rotation of the rotary arm about the axis of rotation causes a corresponding rotation of the extended webbing engagement point about the axis of rotation.
2. The mounting device of claim 1, further comprising a first sensing device configured to capture a first sensor data associated with the intermediate webbing portion.
3. The mounting device of claim 2, further comprising a controller configured to detect a connection condition associated with the fall protection device based at least in part on the first sensor data captured by the first sensing device.
4. The mounting device of claim 1, wherein the rotary arm is configured to contact the intermediate webbing portion at the extended webbing engagement point to at least partially define an arrangement of the webbing relative to the mounting device.
5. The mounting device of claim 4, wherein the extension arm is rigidly secured to the arm base at a proximal arm end, the extension arm defining an arm length that extends from the proximal arm end in an outward radial direction to a distal arm end, wherein the outward radial direction being defined relative to the axis of rotation.
6. The mounting device of claim 5, wherein the rotary arm is configured to facilitate an extended range of motion for an operator relative to the fall protection device.
7. The mounting device of claim 1, wherein the rotary arm is configured for coupling with the fall protection device such that a body of the fall protection device is secured relative to the rotary arm, the rotary arm being configured for engagement with the fall protection device such that a rotation of the rotary arm about the axis of rotation causes a corresponding rotation of the fall protection device through a corresponding range of rotational motion.
8. The mounting device of claim 7, wherein the axis of rotation is defined by a central axis of the arm base, and wherein the rotary arm is configured for securing the body of the fall protection device relative to the arm base of the rotary arm such that the axis of rotation is defined at both the arm base and the fall protection device secured thereto.
9. The mounting device of claim 1, wherein the webbing retention feature defines an opening

configured for the webbing of the fall protection device to be threaded therethrough such that at least a portion of the intermediate webbing portion is disposed within the opening.
10. The mounting device of claim 1, wherein the rotary arm is rotatably attached to an arm interface portion of the mounting bracket defining a downward-facing bottom surface such that the rotary arm is configured to define a position vertically below the mounting bracket upon the mounting bracket being secured relative to the materials handling vehicle, and wherein the axis of rotation is defined in a direction at least substantially perpendicular to the downward-facing bottom surface.
11. The mounting device of claim 2, wherein the first sensor data captured by the first sensing device is configured to facilitate a detection of a movement condition defined by one or more movements of the intermediate webbing portion relative to the webbing retention feature.
12. The mounting device of claim 11, wherein the first sensing device comprises an imaging device and the first sensor data is defined by imaging data comprising at least one image of the intermediate webbing portion.
13. The mounting device of claim 2, wherein the first sensing device is positioned at the webbing retention feature.
14. The mounting device of claim 3, wherein the connection condition associated with the fall protection device is defined by a detection of a connected configuration of the fall protection device relative to an operator attachment operatively secured to an operator.
15. A system comprising: a material handling vehicle comprising a frame element and a platform, wherein the frame element is configured to secure a mounting device for a fall protection device to the material handling vehicle, and wherein the frame element is configured to move vertically with the platform during operation of the material handling vehicle; and the mounting device, wherein the mounting device comprises: a mounting bracket configured for attachment to a materials handling vehicle; and a rotary arm rotatably attached to the mounting bracket at an arm base, wherein the rotary arm is configured to rotate relative to the mounting bracket about an axis of rotation defined at the arm base, wherein the rotary arm defines a webbing retention feature configured for receiving a webbing of the fall protection device to define a dynamic engagement of the rotary arm with an intermediate webbing portion defined along a length of the webbing, the webbing retention feature defining an extended webbing engagement point, and wherein the webbing retention feature is defined at a distal position along an extension arm of the rotary arm, and wherein a rotation of the rotary arm about the axis of rotation causes a corresponding rotation of the extended webbing engagement point about the axis of rotation.
16. The system of claim 15, wherein the fall protection device operably secures an operator wearing a harness to the frame element of the material handling vehicle.
17. The system of claim 16, wherein the fall protection device is secured to one or more anchor points of the harness worn by the operator.
18. The system of claim 15, wherein the frame element is positioned above a workspace defined by the material handling vehicle.